Aerospace series

Plates
Aluminium alloy 7040/7050
High toughness
Close-tolerance flatness
75 mm ≤ a ≤ 220 mm

Dimensions

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Contents

1 Scope

This standard specifies the dimensions, tolerances and mass of 7040/7050 high toughness plates with close tolerance flatness, thickness range 75 ≤ a ≤ 200 mm for aerospace applications.

2 Normative references

This Airbus standard incorporates by dated or undated reference provisions from other publications. All normative references cited at the appropriate places in the text are listed hereafter. For dated references, subsequent amendments to or revisions of any these publications apply to this Airbus standard only when incorporated in it by amendment of revision. For undated references, the latest issue of the publication referred to shall be applied.

AIMS 03-02-000	Technical specification – Aluminium and aluminium alloy wrought products – Plate
AIMS 03-02-019	Material specification – Aluminium alloy (7040/7050) solution treated, controlled stretched and artificially aged (T7451), Plate, 75,0 mm ≤ a ≤ 220,0 mm, high toughness, close tolerance flatness
AIMS 03-02-023	Material specification – Aluminium alloy (7040) solution treated, controlled stretched and artificially aged (T7651), Plate, 120,0 mm ≤ a ≤ 145,0 mm, high toughness, close tolerance flatness

3 Form

See Figure 1.

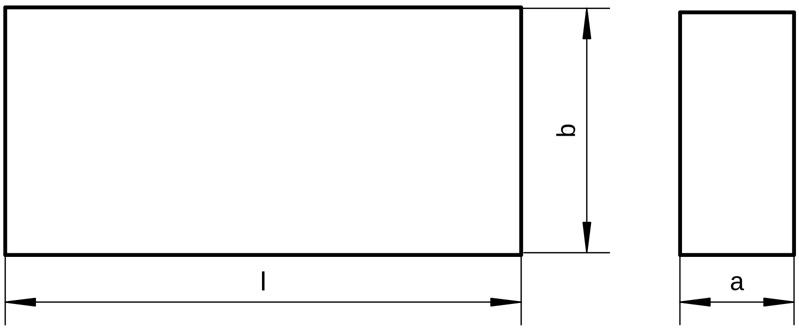

Figure 1:

4 Recommended dimensions

4.1 Dimensions and mass

See Table 1.

Table 1:

Size code	Nominal thickness a [1)] mm	width x length b x l [1)] mm x mm	Mass per unit area [2)] kg/m^2
075	75	1250 x 2500	210
080	80		224
090	90		252
100	100		280
110	110		308
120	120		336
130	130		364
140	140		392
145	145		406
150	150		420
160	160		448
170	170		476
180	180		504
190	190		532
200	200		560
210	210		588
220	220		616

1) Other thickness, width and length on request
2) For information, calculated with a density of 2,8 kg/dm^3

5 Tolerances

5.1 Dimensional tolerances

5.1.1 Thickness

See Table 2.

Table 2:

Dimensions in mm

Thickness a	Thickness tolerances depending on plate thickness a and plate width b				
	b ≤ 1250	1250 < b ≤ 1600	1600 < b ≤ 2000	2000 < b ≤ 2500	2500 < b ≤ 3000
75 ≤ a ≤ 80	± 2,0	± 2,1	± 2,2	± 2,5	± 2,5
80 < a ≤ 100	± 2,2	± 2,2	± 2,4	± 2,7	± 2,7
100 < a ≤ 120	± 2,3	± 2,3	± 2,5	± 2,8	± 2,8
120 < a ≤ 140	± 2,5	± 2,5	± 2,7	± 3,0	± 3,0
140 < a ≤ 160	± 2,6	± 2,6	± 2,8	± 3,1	± 3,1
160 < a ≤ 180	± 2,7	± 2,7	± 2,9	± 3,2	± 3,2
180 < a ≤ 200	± 2,8	± 2,8	± 3,0	± 3,3	± 3,4
200 < a ≤ 220	± 2,9	± 2,9	± 3,1	± 3,4	± 3,6

5.1.2 Width

See Table 3.

Table 3:

Dimensions in mm

Width b	Width tolerances
All	+10 0

5.1.3 Length

See Table 4.

Table 4:

Dimensions in mm

Length l	Length tolerances
l ≤ 5000	+10 0
l > 5000	+0,002 x l 0

5.2 Geometric tolerances

5.2.1 Squareness

5.2.1.1 Method of measurement and symbols

See AIMS 03-02-000.

5.2.1.2 Tolerances

See Table 5.

Table 5:

Dimensions in mm

Length l	Maximum difference in lengths of diagonals for all plate widths and thickness
l ≤ 2000	8
2000 < l ≤ 5000	10
l > 5000	0,002 x l

5.2.2 Lateral curvature

5.2.2.1 Method of measurement and symbols

See AIMS 03-02-000.

5.2.2.2 Tolerances

See Table 6. The lateral curvature may be concave or convex.

Table 6:

Dimensions in mm

Thickness a	Lateral curvature F on	
	plate width b	plate length l
75 ≤ a ≤ 220	≤ 0,002 x b	≤ 0,002 x l

5.2.3 Flatness

5.2.3.1 Method of measurement and symbols

See AIMS 03-02-000.

5.2.3.2 Tolerances

See Table 7.

Table 7:

Dimensions in mm

Plate thickness a	Flatness deviation f_{max}. referred to					
	plate width b		plate length l		chord	
	width	f_{max}	length	f_{max}	chord	f_{max}
75 ≤ a ≤ 220	b ≤ 500 b > 500	1,0 0,002 x min_1 [1)]	l ≤ 500 500 < l ≤ 1000 1000 < l ≤ 2000 l > 2000	1,0 0,002 x l, 2,0 0,001 x min_2 [2)]	w ≥ 300	0,003 x w [3)]

1) min_1 is either plate width (b) or straight-edge length, whatever is the smaller value
2) min_2 is either plate length (l) or straight-edge length, whatever is the smaller value
3) The maximum acceptable deviation for total length and width shall not be exceeded.

6 Material

See Table 8.

Table 8:

Plate thickness range mm	Material					
	delivered			use		
	AIMS spec. no.	condition of treatment	code	AIMS spec. no.	condition of treatment	code
75 ≤ a ≤ 220	03-02-019	T7451	A	03-02-019	T7451	A
120 ≤ a ≤ 145	03-02-023	T7651	B	03-02-023	T7651	B

7 Designation

Material according to this standard shall be designated following the philosophy of the example below:

Description block Identity block

Plate ABS5324 A 080 -01

Number of ABS standard

Material code (see table 8)

Size code (see table 1)

Blank if differentiation between 7040 / 7050 not required
Designation with alloy code only when required by design
Code 01 for 7040
Code 02 for 7050

8 Technical specification

See AIMS03-02-000

RECORD OF REVISIONS

Issue	Clause modified	Description of modification
1 06/00	-	New standard
2 12/00	General	Close tolerance flatness requirements added, maximum thickness extended to 220 mm in line with AIMS 03-02-019 Issue 1 October 2000;
3 06/01	-	Code B added for AIMS03-02-023;
4 04/02	5.1.1 5.2.3.2	Table 2: thickness tolerances for width > 2000 mm added, Table 7: width range modified;
5 02/04	2 and 6	AIMS 03-02-023 title: thickness range corrected, Table 8 amended accordingly
6 06/04	4	Table 1: size code 145 added;
7 01/07	7	Editorial corrections;
8 12/07	5.1.1	Table 2: thickness range 75 ≤ a ≤ 220 mm split into two lines: 75 ≤ a ≤ 80 mm and 80 < a ≤ 100 mm, minimum width (500 mm) removed;
9 05/11	General 2 5.2.1.1, 5.2.2.1 5.2.3.1 5.2.1.2 Table 7 7	Editorial changes according to new template, table column headings reworded, EN 3848 and footnote 1 deleted, Methods of squareness, lateral curvature and flatness measurement refer now to AIMS 03-02-000 (before EN 3848), Squareness tolerance: maximum difference of diagonal lengths has only one limit, Limits revised based on flatness measurement method defined in AIMS 03-02-000 and lower flatness limits introduced, footnotes added, Wording revised and appropriate size code example chosen;